INCH-POUND

A-A-20126C
September 28, 1993
SUPERSEDING
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June 29, 1990

COMMERCIAL ITEM DESCRIPTION

FLOUR

The U.S. Department of Agriculture has authorized the use of this Commercial Item Description in preference to Federal Specifications N-F-481 and N-F-471.

This Commercial Item Description (CID) covers enriched wheat flour, whole wheat flour, medium color rye flour, and oat flour, packed in commercially acceptable containers, suitable for use by the Federal Government.

Salient characteristics.

The product shall conform to the following types and styles, as specified in the solicitation, contract, or purchase order.

Types.

Type I	- Bread Flour (Hard wheat flour)
Type II	- Bread Flour (Bakers hard wheat Hearth-style) (Hearth-style bread flour may contain potassium bromate added in a quantity not exceeding 25 parts to each million parts of finished flour to achieve maximum quality acceptance hearth-style bread products.)
Type III	- General or All Purpose Flour
Type IV	- Cake Flour (Soft wheat flour)
Type V	- Pastry Flour
Type VI	- Whole Wheat Flour
Type VII	- Rye Flour
Type VIII	- Oat Flour

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Head, Food Quality Assurance Staff, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.

FSC 8920

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Styles.

Style 1 - Unbleached (Wheat flour which has not been treated with a maturing ingredient, a color modifying ingredient, or any combination of maturing and color modifying ingredients.)

Style 2 - Bleached (Wheat flour which has been treated with a maturing ingredient, a color modifying ingredient, or any combination of maturing and color modifying ingredients.)

Preparation.

Wheat flour shall be milled and processed in the United States, its territories or possessions, from wheat produced in the United States. The wheat shall be U.S. No. 2 grade or better. Unclassed wheat and mixed wheat are excluded.

The six types of wheat flour shall be milled from wheat of the following classes:

Bread flour, bread flour (Hearth-style), and whole wheat flour	-	Hard red spring Hard red winter White wheat (subclass hard white)
General or All Purpose flour	-	Hard red spring Hard red winter Soft red winter White wheat
Cake flour and pastry flour	-	Soft red winter White wheat (except subclass hard white)

Rye flour shall be milled and processed in the United States, its territories or possessions, from rye produced in the United States. The rye shall be U.S. No. 2 grade or better.

Oat flour shall be milled and processed in the United States, its territories or possessions, from oats produced in the United States. The oat flour shall be food grade, milled from sound, cleaned oats. The oats shall be U.S. No. 2 grade or better.

Processing.

The wheat flour shall be processed through a centrifugal impactor operated at the equipment manufacturer's recommended speed and flow rate designed to insure the destruction of all insect eggs and shall be bolted through a 9XX or finer sieve sifter. Impaction and bolting processes may be repeated at various points in the milling operation. One of the above treatments shall

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occur immediately before entrance of the flour into the packing machinery. The rye flour shall be processed through a centrifugal impactor to destroy insect eggs as necessary, during the processing, but at least immediately prior to entering the packing machinery. The oat flour shall be dehulled oats that have been steamed and ground to produce a stable flour in which enzyme activity has been minimized. Granulation shall not exceed 2 percent on a U.S. Standard No. 35 sieve nor more than 75 percent on a U.S. Standard No. 100 sieve.

Finished product.

The wheat flour shall have a good, characteristic color; shall possess a natural wheat-flour flavor and odor; and shall be free from rancid, bitter, musty, and other undesirable flavors and odors. The wheat flour shall be free flowing and free from lumps. Product shall be suitable for use in baking processes presently in common industrial use for the product(s) as specified by the type of flour and application.

The rye flour shall be a moderately refined product consisting of rye endosperm and small amounts of other parts of the rye kernel. It shall have a good, characteristic, very light, grayish-brown color; shall possess a natural rye-flour flavor and odor; and shall be free from rancid, bitter, musty, and other undesirable flavors and odors. Product shall be suitable for use in combination with other food ingredients for preparation of varietal bakery products.

Oat flour shall have a good, characteristic color; shall possess a natural oat flavor and odor; and shall be free from rancid, bitter, musty, and other undesirable flavors and odors. The oat flour shall be free flowing and free from lumps. Product shall be suitable for use in combination with other food ingredients for preparation of varietal bakery products.

Analytical and physical requirements.

The flour shall conform to the following requirements listed in Table I, Table II, or Table III.

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TABLE I. Analytical and physical requirements for wheat flour 1/

Type	Protein level 2/ (Percent)		Ash level 2/ (Percent)	Moisture (Percent)	FN Units or GAA 3/			
	Min.	Max.	Max.	Max.	Min.	Max.	Min.	Max.
Bread flour 4/	11.3	----	0.48	14.0	225	300	300	600
Bread flour (Hearth-style)	13.3	----	0.55	14.0	200	300	350	600
General or All Purpose flour 5/	9.0	----	0.48	14.0	225	300	300	600
Cake flour 6/	----	9.3	0.40	13.5	350	---	---	---
Pastry flour	----	10.5	0.47	14.0	350	---	---	---
Whole wheat flour	12.0	----	1.90	14.0	250	---	---	---

1/ Analytical procedures shall be performed in accordance with the Approved Methods of the American Association of Cereal Chemists.

2/ Requirements are based on 14.0 percent moisture.

3/ Falling Number (FN) applies to all flours containing diastatic barley malt. Grain Amylase Analyzer (GAA) applies to flours containing fungal alpha-amylase. (1 to 2.5 g fungal alpha-amylase/100 lb. enriched flour.)

4/ Bread flour shall produce bread with a specific volume of not less than 4.8 when tested for specific volume by the rapeseed method.

5/ General or All Purpose flour shall produce a cake with a specific volume of not less than 2.2 without gas pockets when tested for specific volume.

6/ pH level shall be not less than 4.3 minimum nor more than 4.7 maximum.

TABLE II. Analytical and particle size requirements for rye flour 1/

Ash level 2/ (Percent)	Moisture (Percent)	Maximum amount of material permitted on U.S. Standard woven-wire cloth sieve	
		No. 50	No. 100
Maximum	Maximum	(Percent)	(Percent)
1.5	13.5	1.0	20.0

1/ Analytical procedures shall be performed in accordance with the Approved Methods of the American Association of Cereal Chemists.

2/ Requirements are based on 14.0 percent moisture.

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TABLE III. Analytical requirements for oat flour 1/ 2/

Protein level (Percent) Minimum	Fat (Percent) Maximum	Ash (Percent) Maximum	Moisture (Percent) Maximum
13.5	8.0	2.2	10.0

1/ Analytical procedures shall be performed in accordance with the Approved Methods of the American Association of Cereal Chemists.

2/ Requirements are based on 10.0 percent moisture.

Contractor's certification.

By submitting an offer, the contractor certifies that the product offered meets the specified salient characteristics and requirements of this CID; conforms to the producer's own specifications and standards, including product characteristics, manufacturing procedures, quality control procedures, and storage and handling practices; has a national or regional distribution from storage facilities located within the United States, its territories, or possessions; and is sold on the commercial market. The Government reserves the right to determine proof of such conformance prior to the first delivery from point of origin and any time thereafter, up to and including delivery at final destination, as may be necessary to determine conformance with the provisions of the contract.

Regulatory requirements.

The delivered product shall comply with all applicable Federal and State mandatory requirements and regulations relating to the preparation, packaging, labeling, storage, distribution, and sales of the product within the commercial marketplace. Delivered product shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Quality assurance.

When required in the solicitation, contract, or purchase order, the Federal Grain Inspection Service (FGIS), U.S. Department of Agriculture, will determine the quality assurance of the flour according to FGIS procedures. The product shall be examined or analyzed as applicable in accordance with provisions in the CID, and when applicable, the United States Standards for Condition of Food Containers currently in effect on the date of the solicitation.

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and dough oxidizers are stored

Filth test.

All tests required to determine compliance with the provisions of the Federal Food, Drug, and Cosmetic Act in regard to insects, insect fragments, rodent hairs, rodent excreta, and other filth shall be made in accordance with the Official Methods of Analysis of the AOAC or the Approved Methods of the American Association of Cereal Chemists.

Baking quality tests.

For General Purpose Flour - (Cake Formula)

Step 1	Parts by weight
Sugar	43.4
Shortening (Nonemulsified plastic vegetable shortening)	9.5
Salt	0.75
Mono- and diglycerides	0.40
Sodium aluminum phosphate	0.50
Coated monocalcium phosphate	0.05
Step 2	
General purpose flour	41.4
Egg albumen	1.5
Nonfat dried milk	2.0
Bicarbonate of soda	0.5

Mixing Procedure for Cake Mix.

Cream Step 1 ingredients together. Add and mix in Step 2 ingredients. After dry ingredients have been thoroughly blended, add 20 fluid ounces (591.47 mL) of water to 80 ounces (2.27 kg) of blended cake mix and mix for 2 minutes in a mechanical blender on second speed. Add an additional 20 fluid ounces (591.47 mL) of water and mix until a smooth batter is obtained. Before batter is poured into a cake pan, lightly grease the sides of the pan and add a release-type parchment paper to the bottom of the pan. Bake 450 grams of batter at 350°F (176.67°C) for 30 minutes in a round, layer, cake pan having inside dimensions of 8 by 1-1/4 inches (20.32 by 3.175 cm).

For Bread Flour

Procedures shall be performed in accordance with the method in the American Association of Cereal Chemists, as follows:

Test	Number	Method
Baking Quality	10-11	Sponge-Dough, Pound-Loaf

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Specific Volume.

Determination of specific volume for cake and bread shall be accomplished by the rapeseed displacement method. The volume shall be measured after cake or bread cools at least 2 hours in a room or chamber not exceeding 74°F (23.33°C), 55 percent relative humidity.

Remove cooled cake or loaf from pan, place in a container which is higher and greater in perimeter than the cake.

Fill the void space in the container containing the product with rapeseed so that the rapeseed is level with the top edge of the container. Measure this amount of rapeseed in whole milliliters using a graduated cylinder or equivalent measuring device.

Remove product from the container and weigh the product. The weight shall be calculated in terms of whole grams.

Fill the empty container with rapeseed so that the rapeseed is level with the top edge of the container. Measure this amount of rapeseed as specified above.

Calculate specific volume as follows:

$$\frac{\text{Volume of rapeseed in the empty container} - \text{Volume of rapeseed in the void space in the container containing the product}}{\text{Weight of product}} = \text{Specific volume}$$

(Cake volume shall be ascertained by subtracting volume of unoccupied space in container from container volume.)

Preservation, packaging, packing, labeling, and marking.

The flours shall be preserved, packaged, packed, labeled, and cases marked in accordance with good commercial practice. Commercial labeling and packaging, as may be augmented by the solicitation, contract, or purchase order, shall be acceptable. Shipping containers shall comply with the National Motor Freight Classification or Uniform Freight Classification, as applicable.

For Department of Defense procurements (Only).

The following requirements are applicable when specified by the contracting officer.

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A. Additional quality assurance.

Oxygen content test. When specified by the contracting officer, five filled and sealed cans shall be forwarded to an Agricultural Marketing Service (AMS) laboratory for oxygen content testing. The five filled and sealed cans shall be selected at random from the lot. The determination of headspace oxygen for the canned flour shall be by using an electronic oxygen analyzer which operates on the principle of the difference in partial pressure of oxygen between an oxygen reference and the oxygen content of the sample as detected by a porous ceramic Zirconia sensor, such as the Illinois Instruments Analyzer or equivalent; or on the principle of paramagnetic reasonce such as the Servomex Analyzer, or equivalent. The oxygen analyzer shall be calibrated to a known standard prior to testing the headspace gas of the product. Test results shall be reported to the nearest 0.1 percent for each can. The lot shall be rejected if any of the cans exceed 2.0 percent oxygen.

B. Commercial packaging. Same as for commercial practices. (See Preservation, packaging, packing, labeling, and marking.)

C. Export packaging. Ten pounds (4.54 kg) of product shall be packaged in a paper pocket made from heavy duty Kraft. Pockets shall be closed so as to prevent spillage or sifting of product.

Thirty-five pounds (15.88 kg) of Type I, Style 2 product shall be packaged in a 5-gallon (18.927 L), square, metal can - approximate size 9-3/8 by 9-3/8 by 13-3/4 inches (23.81 by 23.81 by 34.925 cm). The can shall have soldered or welded side seams and compound-lined, double-seamed ends. The can top shall have a cover not less than 6-3/16 inches (15.72 cm) in diameter. The sealed can shall not leak.

Fifty pounds (22.68 kg) of product shall be packaged in a multiwall paper shipping sack.

Four pounds, four ounces (1.93 kg) of whole wheat flour or three pounds, four ounces (1.47 kg) of oat flour shall be hermetically sealed in a 603 x 700 can. The can shall be an open-top style, round, metal can with welded side seam and compound-lined ends. The can shall be made throughout from not less than 0.25 pound per base box electrolytic tinplate. The can shall have sufficient base-plate and temper to protect the product during shipment and storage. Prior to sealing, the product shall be gas-packaged by first removing the air from the filled can and replacing it with nitrogen so that the oxygen content of the gas in the filled and sealed container does not exceed 2.0 percent.

D. Export packing. Five 10-pound (4.54 kg) paper pockets shall be packed in a multiwall paper shipping sack.

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One 35-pound (15.88 kg) can of product shall be packed in a fiberboard box, constructed, closed, and reinforced in accordance with good commercial practice.

One 50-pound (22.68 kg) sack, packaged in accordance with export packaging, (see above) shall be acceptable. Alternatively, a commercial package shall be packed in a multiwall, paper, shipping sack.

Six 603 x 700 cans shall be packed in a close-fitting, fiberboard box constructed, closed, and reinforced in accordance with Style RSC, Grade V3c, V3s, or V4s of PPP-B-636. Reinforcement shall be limited to nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape.

E. Unit loads. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for Type II, Class F loads.

F. Labeling. Commercial labeling shall be acceptable.

G. Marking. Marking of shipping containers and unit loads shall be in accordance with MIL-STD-129.

Notes.

Purchasers should specify:

- Type(s) and style(s) of flour.
- If flour is to be bromated (Type II only).
- When baking quality tests are to be performed.
- Bulk shipment.

Sources of documents.

Sources of information for nongovernmental documents are as follows:

Copies of the National Motor Freight Classification may be obtained from:

National Motor Freight Traffic Association, Inc., Agent
National Motor Freight Classification
American Trucking Associations, Inc., Traffic Department
2200 Mill Road
Alexandria, VA 22314

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Copies of the Uniform Freight Classification may be obtained from:

Uniform Classification Committee, Agent
Uniform Freight Classification
Uniform Classification Committee, Suite 1120
222 South Riverside Plaza
Chicago, IL 60606

Copies of the Official Methods of Analysis of the AOAC may be obtained from:

AOAC International
2200 Wilson Boulevard
Suite 400
Arlington, VA 22201-3301

Copies of the Approved Methods of the American Association of Cereal Chemists may be obtained from:

American Association of Cereal Chemists
3340 Pilot Knob Road
Saint Paul, MN 55121

Sources of information for governmental documents are as follows:

Applicable provisions of the Federal Food, Drug, and Cosmetic Act are contained in 21 CFR Parts 1-199. This three-volume set may be purchased from:

Superintendent of Documents
U.S. Government Printing Office
Washington, DC 20402-0001

Credit card (MasterCard or Visa) purchases may be made by calling the Superintendent of Documents on (202) 783-3238.

Copies of the United States Standards for Condition of Food Containers are available from:

Chairperson
Condition of Container Committee
Agricultural Marketing Service
U.S. Department of Agriculture
Room 2506, South Building
P.O. Box 96456
Washington, DC 20090-6456

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Copies of the Official U.S. Standards for Grain are available from:

Federal Grain Inspection Service
Animal and Plant Health Inspection Service
Printing and Distribution Section
G-100 Federal Building
Hyattsville, MD 20782

Civil agencies and other interested parties may obtain copies of this CID from:

General Services Administration
Specifications Unit (3FB-WS)
Room 6654
7th and D Streets, SW
Washington, DC 20407

Military activities should submit requests for copies of this CID to:

Standardization Documents Order Desk
Building 4, Section D
700 Robbins Avenue
Philadelphia, PA 19111-5094

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Navy - SA
Air Force - 50

Review Activities

Army - MD, QM
Navy - MC
DLA - SS

CIVIL AGENCY COORDINATING ACTIVITIES:

DOJ - BOP
HHS - NIH, IHS
USDA - FV
VA - OSS

PREPARING ACTIVITY:

USDA - FV

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